

## Read Me

The following instruction lists code and data to duplicate the results in the paper “Dominated Options in Health Insurance Plans” by Chenyuan Liu and Justin Sydnor: **openicpsr-124622**. The main data for the project come from the Kaiser Family Foundation and we are not authorized to post them publicly. We give instruction on how to obtain these data.

## Data Application Instruction

1. The KFF data can be obtained by submitting an online form via <https://www.kff.org/contact-us/> under the “TOPIC: Health Costs.” You request the Kaiser Family Foundation Employer Health Benefits Survey (KFF EHBS) data from 2011 through 2016.
2. We use the 2014 Medical Expenditure Panel Survey Full Year Consolidated file, obtained via [https://meps.ahrq.gov/data\\_stats/download\\_data\\_files\\_detail.jsp?cboPufNumber=HC-171](https://meps.ahrq.gov/data_stats/download_data_files_detail.jsp?cboPufNumber=HC-171) . We include the data extract we used in this paper in the code pack: /MEPS/H171.dta.
3. The AV Calculator is downloaded from <https://wayback.archive-it.org/2744/20200125161105/https://www.cms.gov/CCIIO/Resources/Regulations-and-Guidance/Downloads/Final-2016-AV-Calculator-011514.xlsm> We include in this code pack a modified version with extra code used to generate simplified plan designs as described in the paper. This modified file is available in the code pack: /GetSimplifiedPlan/GetSimplifiedPlan.xlsm. We also include in the code pack a copy of the CMS continuance table: riskdist.xlsx in /Analysis/.

## Software Requirements

- Stata
- Matlab
- R
  - tidyverse, lfe packages needed
- Excel

## Time Requirements

- Sometimes Excel gets stuck calculating the AV. To avoid that, before running `anormal()` and the subsequent macros, try inputting some random plan information in the AV calculator and calculating AV manually for this plan.
- Sometimes Excel gets stuck or becomes very slow running the macro `agetsimplenormal()`. One solution is to split the dataset and run the macro piece by piece. This could be achieved by manually changing the loop index.

### **/Table1/**

This folder contains code to generate **Table 1**, summary statistics of plans offered in 2011 to 2016. The data, which we labeled as "health\_benefits\_t.csv", (t = 2011 – 2016) can be obtained from KFF and is not included in the folder.

1. Run ExtractKFF\_2011\_2016.R
2. Run table1.do

### **/GetSimplifiedPlan/**

The folder contains data and code to create the main sample in our analysis and calculate the simplified three-arm design. The KFF data for 2015 in STATA format ("health\_benefits\_2015.dta") was created from the KFF .csv files and is not included in the folder. The AV Calculator created by CMS and used in calculating the simplified three-arm design is attached.

1. Run clean\_2015.do
2. Open GetSimplifiedPlan.xlsm, enable Macro. Go to Developer -> View Code -> Modules -> Module16. Double click to show the code. This file contains the data and program developed by CMS to calculate AV of a plan, as well as author-written program to calculate AV and generate the equivalent coinsurance rate for plans in our sample.
3. Run the macro aprepave()
4. Run the following macros in sequence
  - a. anormal()
  - b. abothcc()
  - c. aggregate()
  - d. agetsimplenormal()

### **Note**

1. Sometimes Excel gets stuck calculating the AV. To avoid that, before running anormal() and the subsequent macros, try inputting some random plan information in the AV calculator and calculating AV manually for this plan.
2. Sometimes Excel gets stuck or becomes very slow running the macro agetsimplenormal(). One solution is to split the dataset and run the macro piece by piece. This could be achieved by manually changing the loop index.

### **/MEPS/**

This folder contains code and data to generate the MEPS distributions we used in robustness analysis in Table 4.

1. Run meps.do
2. Copy information from MEPSdist\_2.smcl to risk\_dist\_meps2.xlsx.

## **/Analysis/**

This folder contains code to generate the main analysis results in the paper.

1. avcom.m: main code containing results in **Figure 1-3, Appendix Figure 1 and Table 2-6**.
  - a. To make sure the main program can run, put the following self-defined MATLAB functions in the same folder as avcom.m: filldist\_hsa.m, filldist\_hsainvest1.m, filldist5.m, fillmeps.m, oop.m, oop2.m, oph.m, opinv1.m.
  - b. avcom.m also imports the following data, so they also need to be put in the same folder as avcom.m: riskdist.xlsx, riskdist\_meps2.xlsx, runavc.xlsx.
  - c. avcom.m only replicates 2 columns in Table 2: “Number of Firms” and “Fraction of Firms”. The numbers in Column “Avg Enrollment Share in HD Plan” is created in dominance.do.
2. dominance.do: generate **Table 2, Table 7, Appendix Table B1, results in Appendix A (Analyses with Family Coverage), and Appendix Table C2 (last two columns)**.
  - a. The program uses data "health benefits 2015.dta" (not included) and runavc.xlsx (output from /GetSimplifiedPlan/).

## **/AppendixC/**

This folder contains code to generate **Appendix Table C1 and C2 (first two columns)**.

1. Run AppendixC\_export.do
2. Open GetSimplifiedPlan.xlsm, enable Macro. Go to Developer -> View Code -> Modules -> Module16. Double click to show the code. [note: may need to change folder path to make sure the input takes in results from AppendixC.do, and the output is in the right folder.]
3. Run the macro aprepare()
4. Click cell B55 in the worksheet “AV Calculator”. Click the button in the pop-up window.
5. Run the following macros in sequence
  - a. anormal()
  - b. abothcc()
  - c. aggregate()
6. Run AppendixC\_analysis.do